

4 A Initial velocities for both stones, $u = 0$

$$s = ut + \frac{1}{2}gt^2 = \frac{1}{2}gt^2$$

For the same time interval, Δt , both stones fall down by the same distance, s .

Therefore the distance between them will always remain the same as that at the point of release.

5 D $F = M \cdot A = (95)(999)(9.4 \cdot 10^{-6})(9.8)^2 = 8.997 \text{ N}$